

Self-check pattern

Before scoreboards and agents, every testbench needs a self-check loop

Stimulus, expect, compare

- Starter: a two-input AND gate with a equals one and b equals one
- The expect comes from the AND truth table, not from copying the DUT output back
- Change one input and recompute expect before you compare
- Wrong expect on purpose should fail so you practice the error path
- Compare after the signal settles, not mid-transition

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Self-checking TB pattern

Pre-UVM literacy: **stimulus** → DUT → **expect** → compare → **pass/fail** (no eyeballing waves alone). Starter: and2 with a=1, b=1, expect y=1 — PASS.

Starter example: and2 with a=1, b=1, expect y=1, already checked — **PASS**.

Load starter example

Challenges 0 / 22 cleared

Quiz: goal: A self-checking TB's main job is...

☐ compare DUT outputs to an expected value and report pass/fail

☐ only dump VCD for manual review

☐ synthesize the DUT

☐ replace Accellera UVM

Show hint

Check

Next

Idle

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Core ideas

Stimulus

Drive DUT inputs from the TB.

Expect

Golden value from a reference model.

Compare

got !== exp → \$error (fail).

Tally

Count errors; exit with a clear result.

Browser lab starter

learn_cocotb — self check tb

Real cocotb practice

- In the real cocotb track, restate the loop on paper before you write a live assert
- Draw four boxes: stimulus, reference model, DUT sample, compare
- Write one cocotb-style check in pseudocode
- Optional: open the lab's source sketch and read the commented bench template
- This module is compare literacy, the scoreboard queue comes from the same expect idea

Pitfalls to watch

- Do not treat no error as pass if you never compared anything
- Do not echo the DUT output into expect, that hides bugs
- A common miss is comparing before the simulator has settled after a poke
- Another trap is only checking in waves
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and predict pass before you run check
- On paper, write the four self-check steps for one input change on the AND gate
- When you are ready